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(54) **GEAR IDENTIFICATION TOOL FOR VEHICLE PARTS**

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4,831,872 A * 5/1989 Huang G01M 13/021
702/167
5,237,862 A * 8/1993 Mangrulkar G01M 15/11
73/114.04
5,497,844 A * 3/1996 Fritzinger A63G 25/00
74/462
5,513,442 A * 5/1996 Flair G01M 13/021
33/501.13
6,040,910 A * 3/2000 Wu G01B 11/2527
356/613
6,711,855 B1 * 3/2004 Daniels E05F 15/63
49/342
9,945,447 B2 * 4/2018 Telep F16H 57/082
2011/0030489 A1 * 2/2011 Chen F16H 55/18
74/440
2013/0133168 A1 * 5/2013 Yamaguchi G01B 11/2416
29/284
2015/0081083 A1 * 3/2015 Kreschel G05B 19/182
33/501.7

(Continued)

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FOREIGN PATENT DOCUMENTS

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CN 115342713 A * 11/2022 G01B 5/0002
FR 1322503 3/1963

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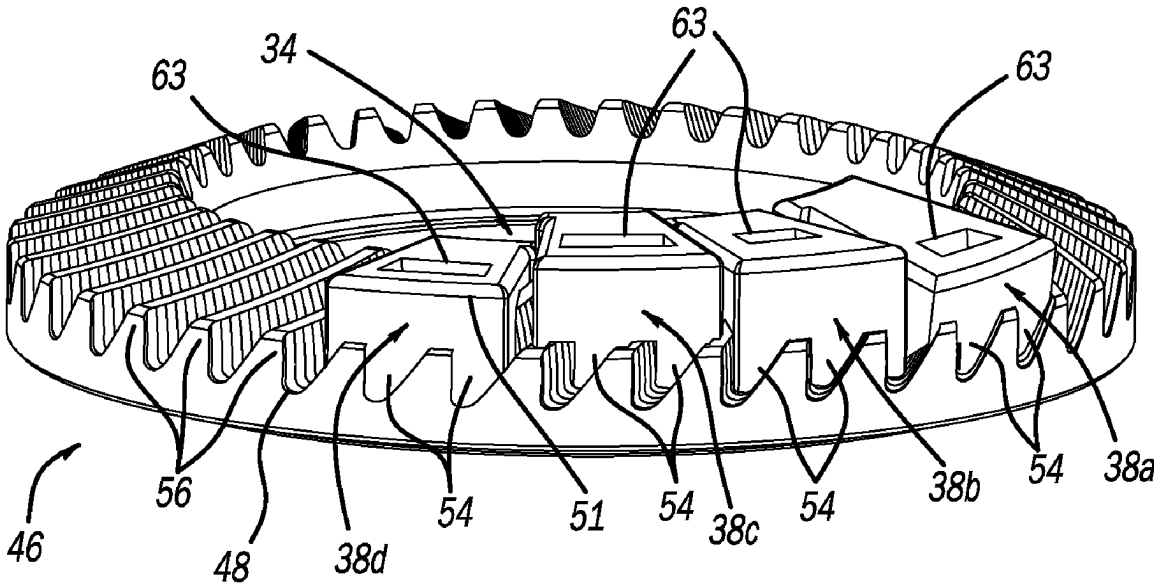
(57) **ABSTRACT**

A gear identification tool includes a segmented assembly including a first body and a second body movable relative to each other. The first and second bodies are coupled to each other by an attachment structure. The first body includes a plurality of first gear teeth configured to mesh with first predetermined gear teeth of a first gear. The first gear teeth have a first teeth profile. The second body includes a plurality of second gear teeth configured to mesh with second predetermined gear teeth of a second gear. The second gear teeth have a second teeth profile that is different than the first teeth profile.

(56) **References Cited**
U.S. PATENT DOCUMENTS

20 Claims, 8 Drawing Sheets

3,167,867 A 2/1965 Reef
4,261,198 A * 4/1981 Moore G01M 13/021
73/162



(56)

References Cited

U.S. PATENT DOCUMENTS

2015/0143890	A1 *	5/2015	Meduru	G01M 13/021
				73/162
2015/0338201	A1 *	11/2015	Wuerfel	B23F 23/006
				33/504
2017/0167589	A1 *	6/2017	Berlinger, Jr.	F16H 1/06
2020/0368835	A1 *	11/2020	Huber	G05B 19/4065
2022/0003177	A1 *	1/2022	Joseph	F02D 41/009

* cited by examiner

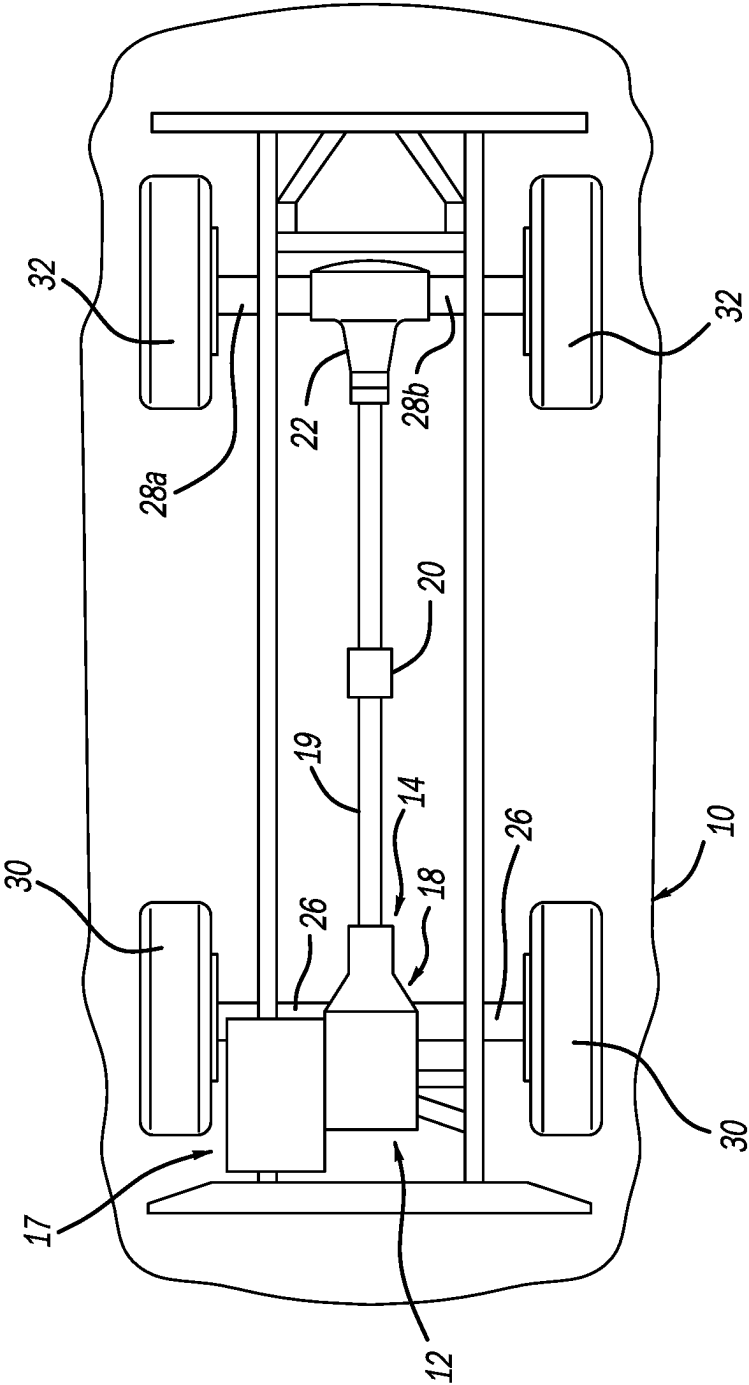


FIG. 1

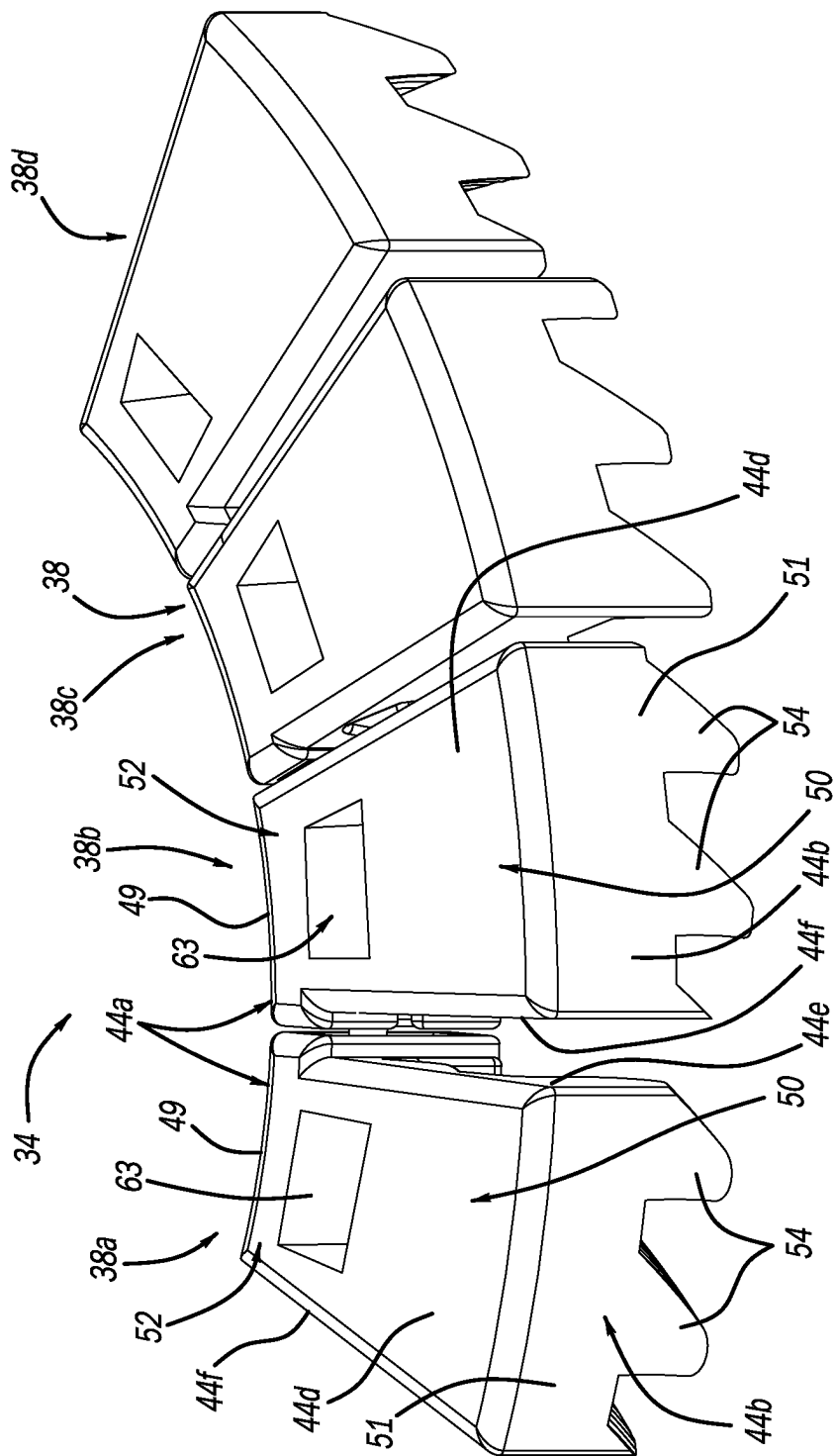


FIG. 2

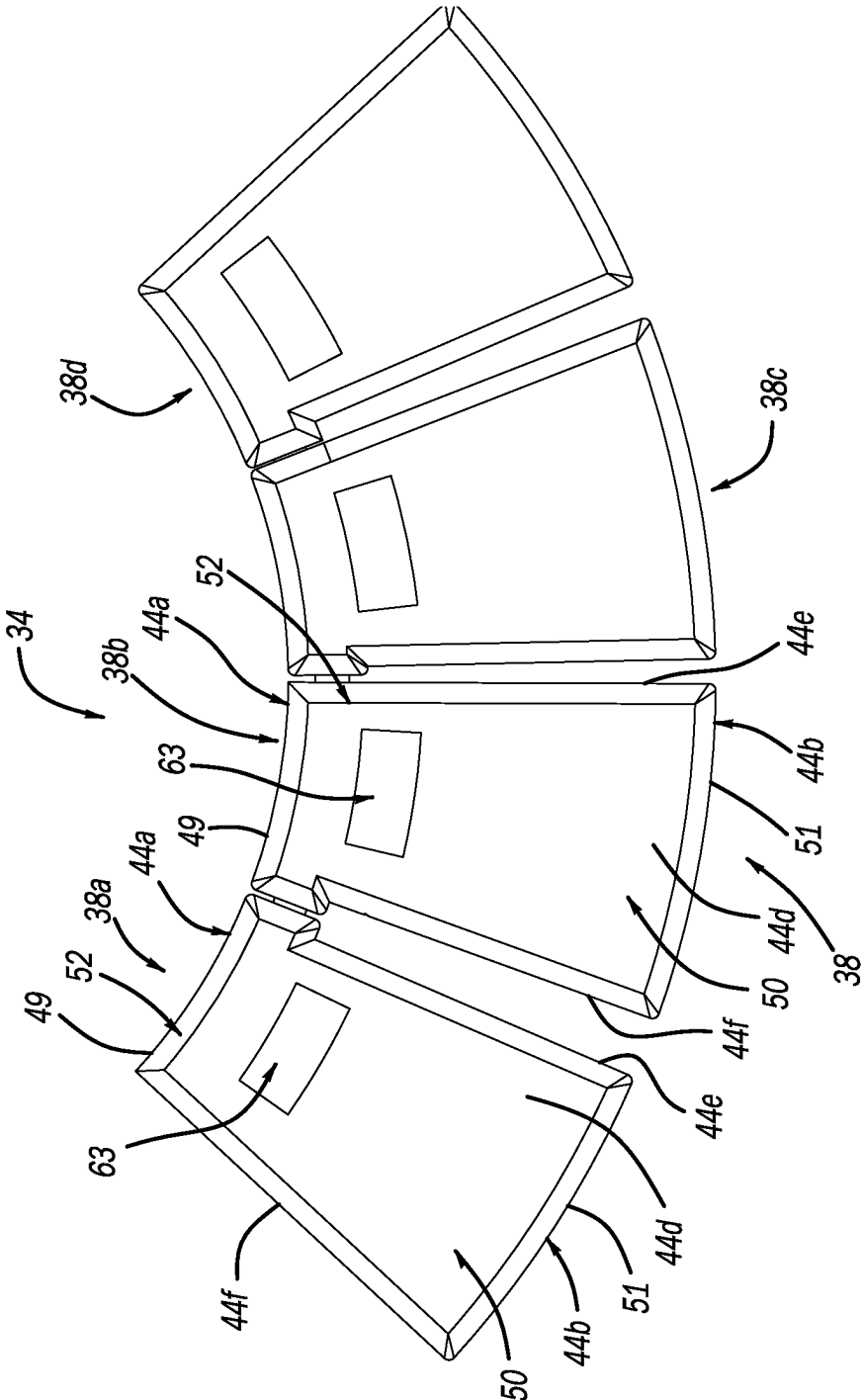


FIG. 3

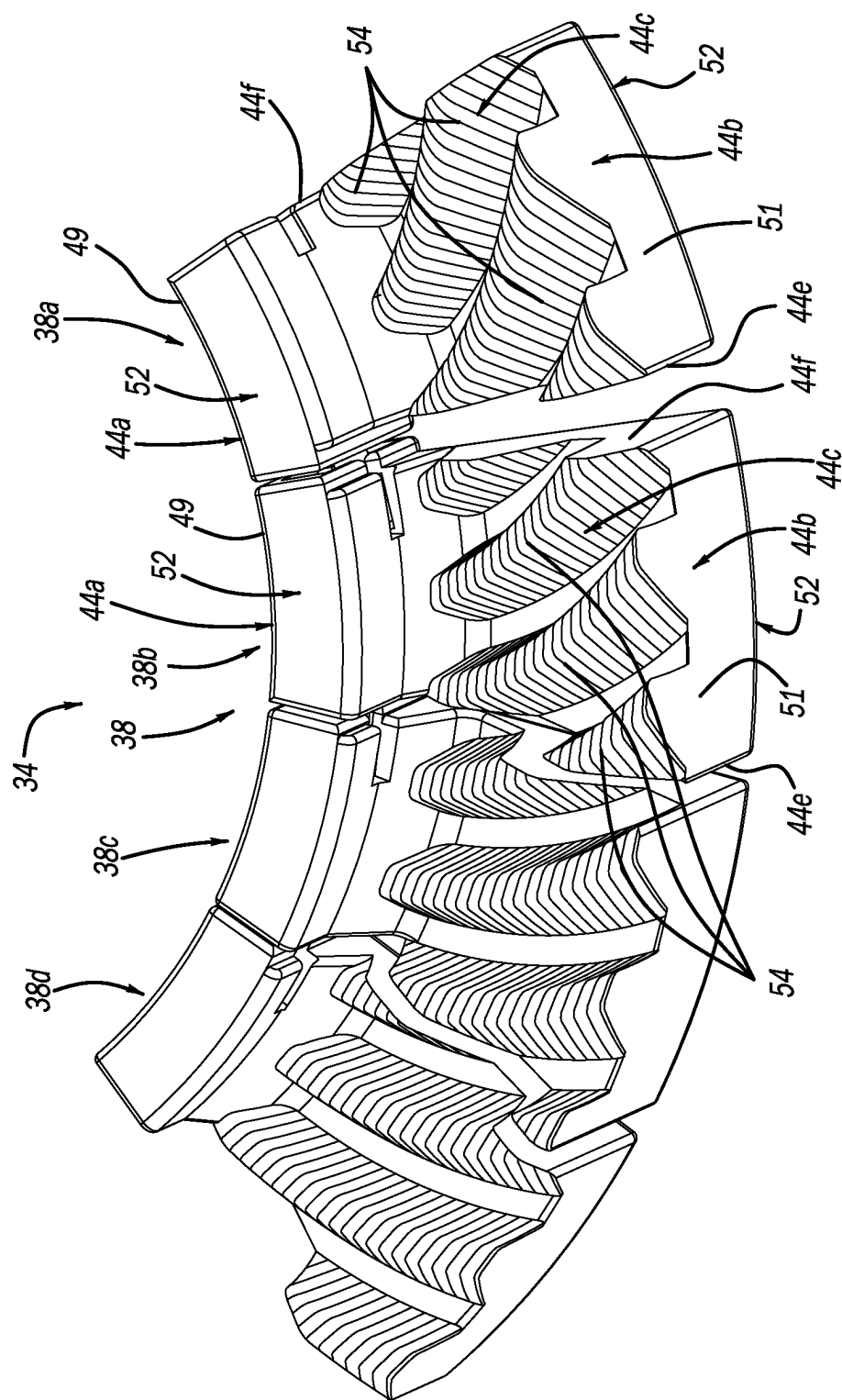


FIG. 4

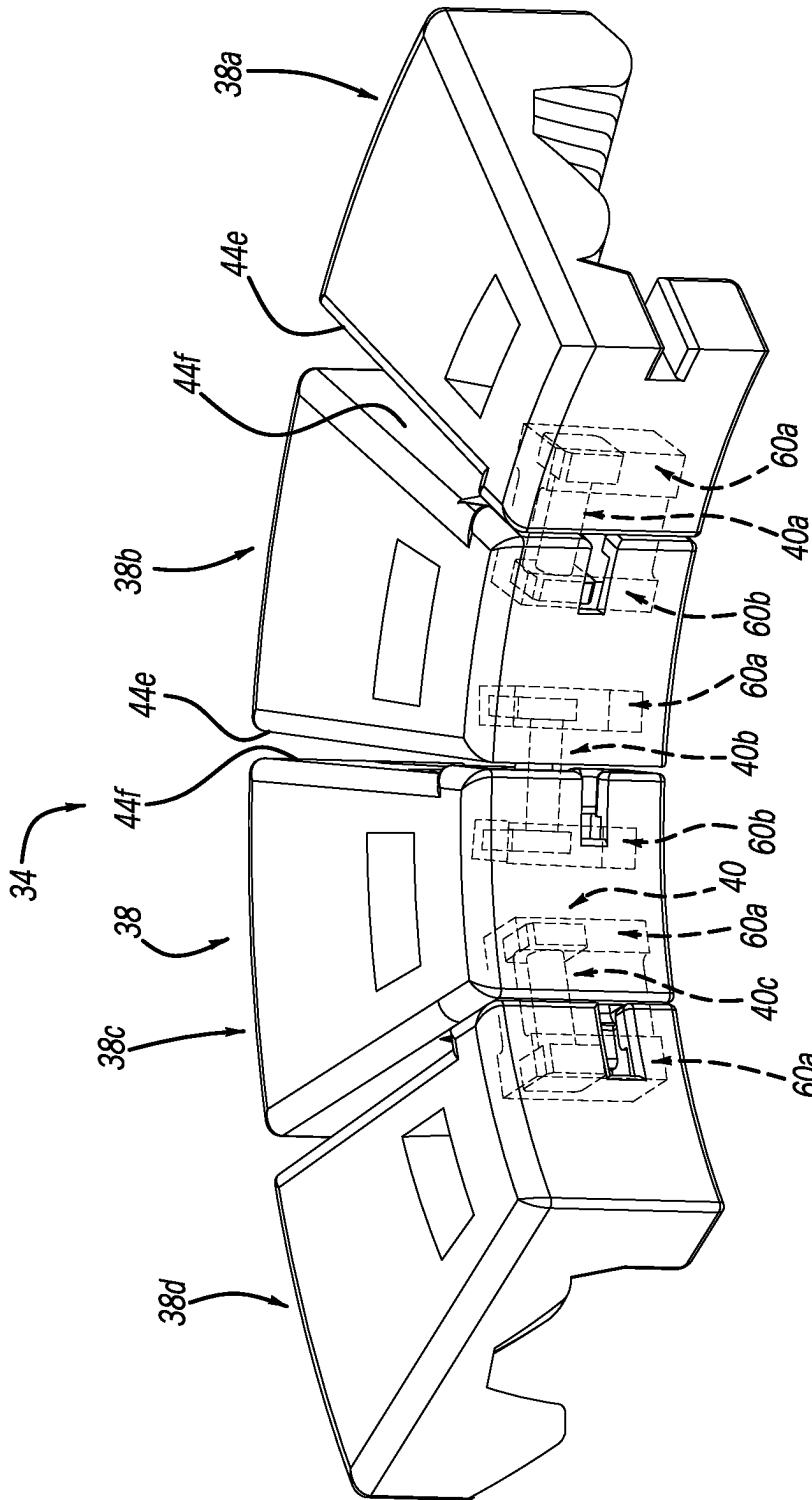
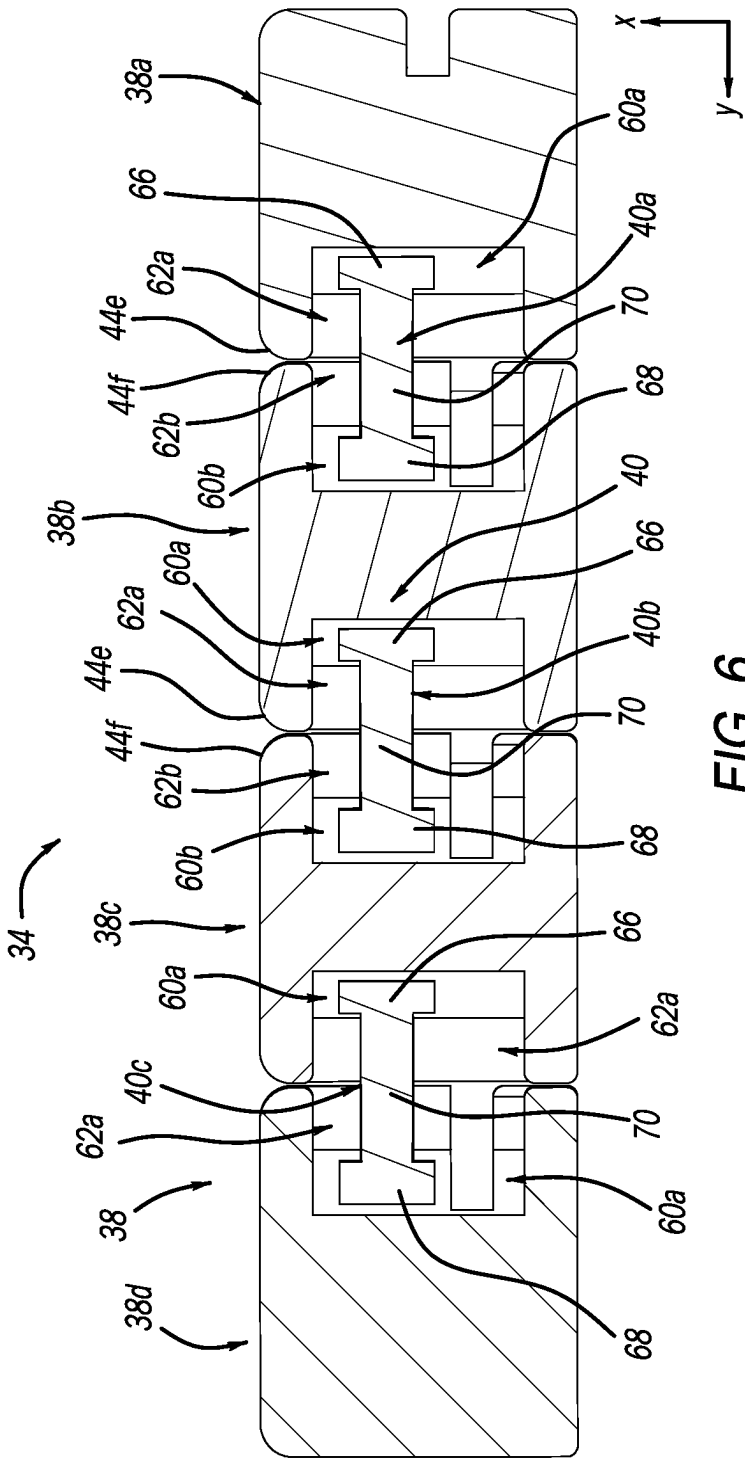


FIG. 5



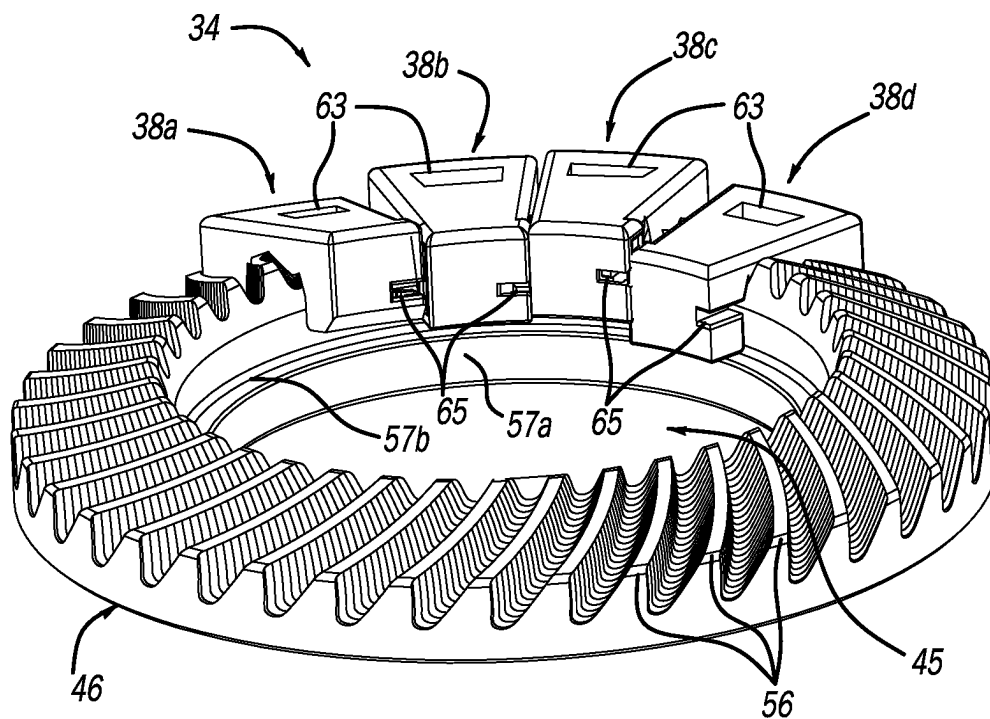


FIG. 7A

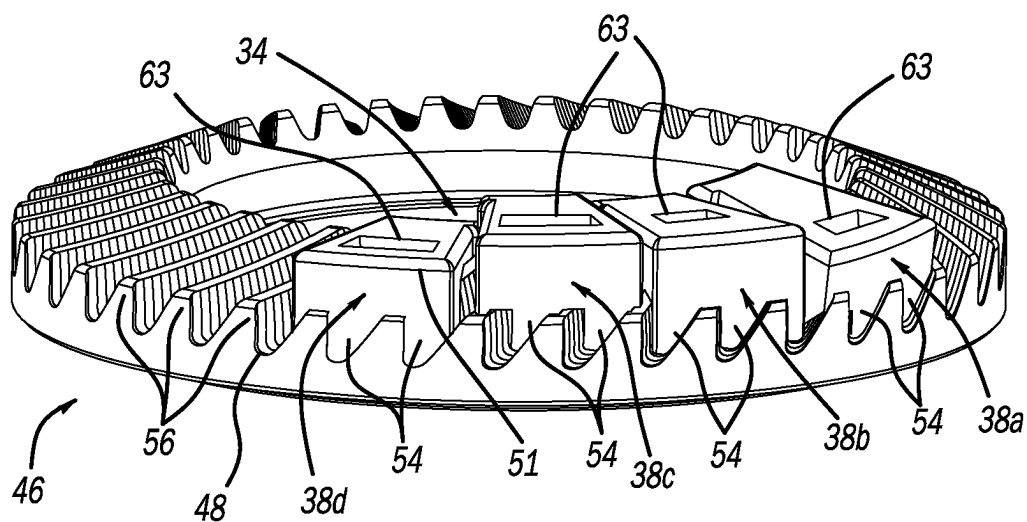


FIG. 7B

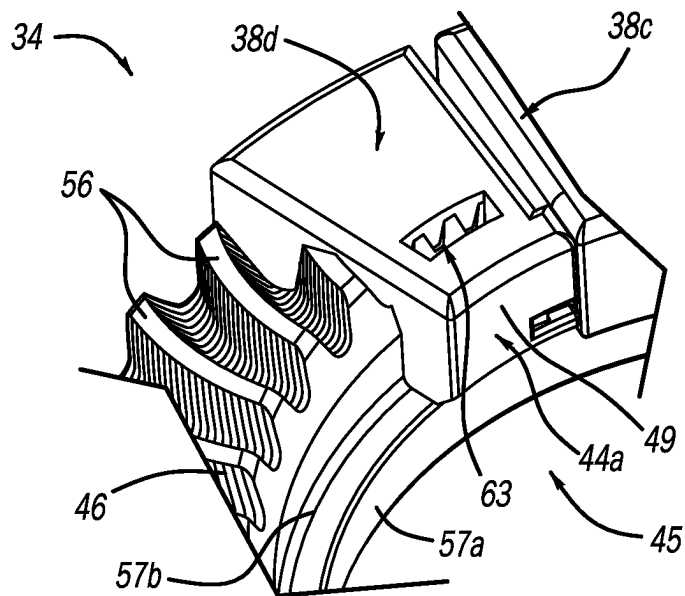


FIG. 7C

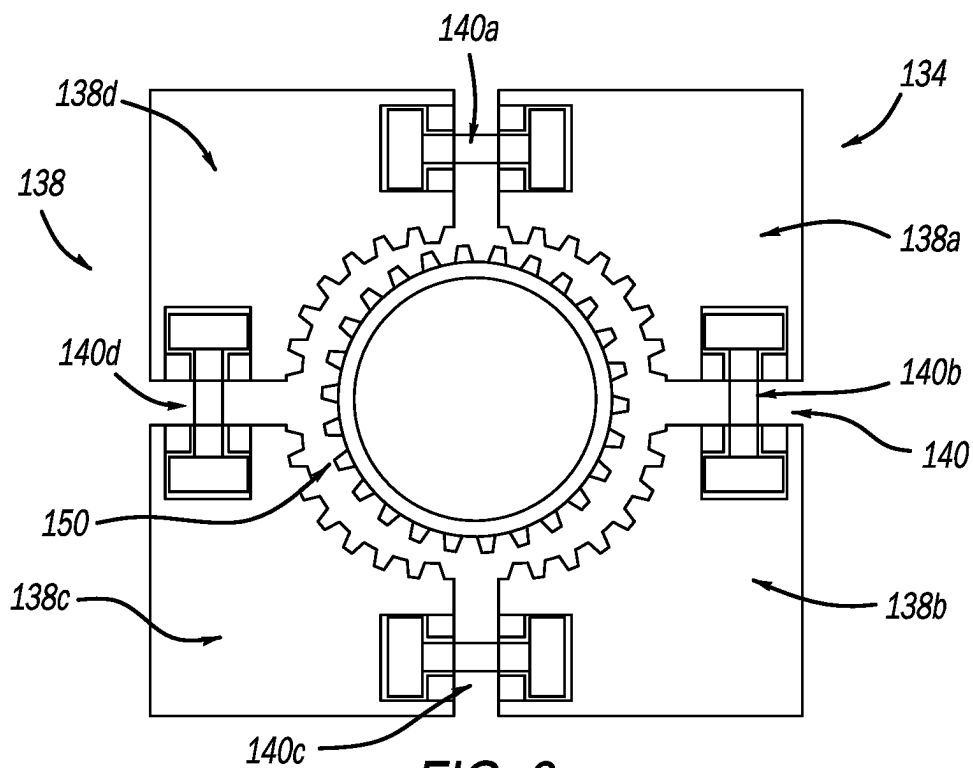


FIG. 8

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GEAR IDENTIFICATION TOOL FOR VEHICLE PARTS

FIELD

The present disclosure relates to a gear identification tool for a part, and more specifically, a gear identification tool for a gear.

BACKGROUND

The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

During the manufacturing process of vehicle parts (e.g., axle gears, transmission gears, splined shafts), similar variations of vehicle parts may be produced in the same production line, for example, with varying levels of completeness, which can make it challenging for an operator of the line to identify the parts from each other. In some forms, teeth of the vehicle parts are counted to identify the vehicle parts from each other. However, counting teeth of vehicle parts can be a lengthy process as some vehicle parts have a large number of teeth (e.g., some vehicle parts contain 40 or more teeth). Furthermore, some vehicle parts have the same number of teeth, which may cause confusion among the line operators.

The teachings of the present disclosure address these and other issues with identifying vehicle parts during the manufacturing process.

SUMMARY

This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

In one form, the present disclosure provides a gear identification tool that includes a segmented assembly including a first body and a second body movable relative to each other. The first and second bodies are coupled to each other by a first attachment structure. The first body includes a plurality of first gear teeth configured to mesh with first predetermined gear teeth of a first gear. The first gear teeth have a first teeth profile. The second body includes a plurality of second gear teeth configured to mesh with second predetermined gear teeth of a second gear. The second gear teeth have a second teeth profile that is different than the first teeth profile.

In variations of the gear identification tool of the above paragraph, which can be implemented individually or in any combination: the first body defines a first internal slot and a first aperture in communication with the first internal slot, the first aperture opens through a first side of the first body; the second body defines a second internal slot and a second aperture in communication with the second internal slot, the second aperture opens through a second side of the second body; the first attachment structure extends between the first side of the first body and the second side of the second body; the first attachment structure permits the first and second bodies to move in a first direction relative to each other and inhibits the first and second bodies from moving in a second direction relative to each other, the second direction being perpendicular to the first direction; the first attachment structure permits the first and second bodies to move within a predetermined distance relative to each other; the first attachment structure includes a first nut located in a first internal slot of the first body, a second nut located in a

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second internal slot of the second body, and a shaft secured to the first and second nuts; the first attachment structure is a unitized structure; the segmented assembly includes a third body coupled to the second body by a second attachment structure, the third body is permitted to move relative to the first and second bodies; the third body includes a plurality of third gear teeth configured to mesh with third predetermined gear teeth of a third gear, the third gear teeth having a third teeth profile that is different than the first and second teeth profiles; the first body has a first side including the first gear teeth and a second side opposite the first side, the second side comprising a window that provides access to the first teeth; the segmented assembly has an arcuate profile disposed about a central axis, the first body extending a first angle relative to the central axis and the second body extending a second angle relative to the central axis; and the first and second gear teeth are face gear teeth.

In another form, the present disclosure provides a gear identification tool that includes a segmented assembly including a plurality of bodies coupled to each other by attachment structures and movable relative to each other. Each body includes a plurality of gear teeth configured to mesh with predetermined gear teeth of a respective gear. Each body includes an internal slot and an aperture in communication with the internal slot. The aperture opens through a first side of the body. One attachment structure of the attachment structures includes a first nut located in the internal slot of one body of the plurality of bodies, a second nut located in the internal slot of another body of the plurality of bodies, and a shaft secured to the first and second nuts.

In variations of the gear identification tool of the above paragraph, which can be implemented individually or in any combination: the gear teeth of the one body of the plurality of bodies have a first teeth profile and the gear teeth of the another body of the plurality of bodies have a second teeth profile, the first gear teeth profile is different than the second teeth profile; the one attachment structure permits the one body and the another body to move within a predetermined distance relative to each other; the one attachment structure permits the one body and the another body to move in a first direction relative to each other and inhibits the one body and the another body from moving in a second direction relative to each other, the second direction being perpendicular to the first direction; the one body has a second side including the gear teeth and a third side opposite the second side, the third side comprises a window that provides access to the gear teeth; the segmented assembly has an arcuate profile disposed about a central axis, the one body extending a first angle relative to the central axis and the another body extending a second angle relative to the central axis; and the segmented assembly includes at least four bodies.

In yet another form, the present disclosure provides a gear identification tool that includes a segmented assembly that includes a plurality of bodies and attachment structures. The plurality of bodies are movable relative to each other. Each body includes a plurality of gear teeth having a gear teeth profile that is different from the gear teeth profile of the other bodies of the plurality of bodies. Each body also includes an internal slot and an aperture in communication with the internal slot. The aperture opens through a first side of the body. The attachment structures couple the plurality of bodies to each other. One attachment structure of the attachment structures includes a first nut located in the internal slot of one body of the plurality of bodies, a second nut located in the internal slot of another body of the plurality of bodies, and a shaft secured to the first and second nuts. The one

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attachment structure permits the one body and the another body to move in an axial direction relative to each other and inhibits the one body and the another body from moving in a radial direction relative to each other. Each of the plurality of bodies has a second side including the gear teeth and a third side opposite the second side. The third side includes a window that provides access to the gear teeth.

Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

DRAWINGS

In order that the disclosure may be well understood, there will now be described various forms thereof, given by way of example, reference being made to the accompanying drawings, in which:

FIG. 1 is a bottom view of a vehicle having a transmission and axle shafts assembled according to the principles of the present disclosure;

FIG. 2 is a perspective view of a gear identification tool according to the principles of the present disclosure;

FIG. 3 is a top view of the gear identification tool of FIG. 2;

FIG. 4 is a perspective view of the gear identification tool of FIG. 2;

FIG. 5 is a perspective view of a portion of the gear identification tool of FIG. 2 with attachment structures shown in hidden lines;

FIG. 6 is a cross-sectional view of a portion of the gear identification tool of FIG. 2;

FIG. 7A is a perspective view of the gear identification tool of FIG. 2 disposed on a gear component according to the principles of the present disclosure;

FIG. 7B is another perspective view of the gear identification tool of FIG. 2 disposed on the gear component according to the principles of the present disclosure;

FIG. 7C is yet another perspective view of a portion of the gear identification tool of FIG. 2 according to the principles of the present disclosure; and

FIG. 8 is a top view of another gear identification tool according to the principles of the present disclosure.

The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

DETAILED DESCRIPTION

The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.

With reference to FIG. 1, a vehicle 10 is provided that includes a powertrain system 12 and a drivetrain system 14. The systems 12, 14 can be conventional systems, for example. In the particular configuration shown in FIG. 1, the powertrain system 12 includes, inter alia, an engine 17 and a transmission 18. The drivetrain system 14 includes, inter alia, a propeller shaft 19, a center differential 20, a rear differential 22, a pair of front shafts or axles 26, and a pair of rear shafts or axles 28a, 28b. Rotary power generated by the powertrain system 12 is transmitted to the drivetrain system 14. That is, rotatory power generated by the powertrain system 12 is transmitted to the pair of front axles 26

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to drive a set of front wheels 30. Similarly, rotatory power generated by the powertrain system 12 is transmitted to the pair of rear axles 28a, 28b (via the propeller shaft 19, the center differential 20 and the rear differential 22) to drive a set of rear wheels 32. In some forms, the vehicle 10 can be an electric vehicle such as a battery electric vehicle (BEV), wherein drive wheels (not shown) are driven by electric motors (not shown) that receive power from a battery pack (not shown). In other examples, the vehicle 10 may be a hybrid electric vehicle (HEV), a plug-in electric vehicle (PHEV), or a fuel cell vehicle, among others.

The differential 22 may be any suitable type of differential. In one form, the differential 22 has a planetary differential gearset and include, inter alia, a driveshaft (not shown), a housing (not shown), a ring gear (not shown), one or more planet gears (not shown), first and second side gears (not shown), and a clutch (not shown). An input gear (not shown) is connected to the driveshaft. The input gear is a separate component that is secured to the driveshaft or may be an integral part of the driveshaft. The input gear is configured to mesh with the ring gear. The planet gears are connected to the ring gear and meshes with the first and second side gears. The first side gear is connected to the rear shaft 28a and the second side gear is connected to the rear shaft 28b. The first side gear is a separate component that is secured to the shaft 28a or may be an integral part of the first shaft 28a. Similarly, the second side gear is a separate component that is secured to the second shaft 28b or may be an integral part of the second shaft 28b.

The clutch (not shown) can be any suitable type of clutch that is operable to selectively permit or inhibit relative rotation between the first and second shafts 28a, 28b. In one form, the clutch (not shown) includes a set of plates (not shown) associated with (e.g., secured to) the housing (not shown) and a set of discs (not shown) associated with (e.g., secured to) at least one of the first and second shafts 28a, 28b. When the set of plates and the set of discs are disengaged from each other, the differential 22 is in a first differential mode and acts as an open differential (i.e., the shafts 28a, 28b are allowed to rotate at different speeds relative to each other). When the set of plates and the set of discs are engaged with each other, the differential 22 is in the second differential mode and acts as a limited slip differential (i.e., the differential 22 inhibits relative rotation between the shafts 28a, 28b). It should be understood that although the differential 22 is described above as a clutch-type limited slip differential, the differential 22 may be other suitable differentials.

With reference to FIGS. 2-6, a gear identification tool 34 is provided for identifying different vehicle parts or components having teeth, for example. For example, during the manufacturing process, similar variations of vehicle parts (e.g., axle gears, transmission gears, splined shafts) may be produced in the same production line, for example, with varying levels of completeness, which can make it challenging for an operator of the line to identify the parts from each other. The gear identification tool 34 may assist the operator in identifying and sorting a plurality of different vehicle parts from the same production line.

The gear identification tool 34 can be made out of a metal material, a plastic material, a composite material, an elastomeric material (e.g., rubber), or any other suitable material. The gear identification tool 34 can be manufactured by an additive manufacturing process (3D printing). The manufacturing process can include laser sintering, for example, that can generally include a laser, a means for applying subsequent layers of powdered sintering material (e.g.,

metal powder), and a controller that controls operation of the laser and the amount and timing of the deposition of the powder. It should be understood that other 3D printing/additive manufacturing methods may be employed, along with a variety of different materials, while remaining within the scope of the present disclosure.

In the example illustrated, the gear identification tool **34** is a segmented assembly having an arcuate profile and including a plurality of bodies **38a**, **38b**, **38c**, **38d** (together referred to as the bodies **38**) and attachment structures **40a**, **40b**, **40c** (FIGS. 5 and 6; together referred to as the attachment structures **40**). In some forms, the segmented assembly **36** may have a linear profile, a circular profile, or any other suitable profile that corresponds to the profile of the parts being identified. The bodies **38** are coupled to each other by the attachment structures **40** and are movable relative to each other as will be described in more detail below. In the example illustrated, the gear identification tool **34** includes four bodies **38**. In some forms, the gear identification tool **34** may include more than four bodies **38** or less than four bodies **38** without departing from the scope of the present disclosure.

Each body **38a**, **38b**, **38c**, **38d** of the gear identification tool **34** includes a plurality of sides **44a**, **44b**, **44c**, **44d**, **44e**, **44f**. When the gear identification tool **34** is disposed on the vehicle part **46**, the side **44a** of each body **38a**, **38b**, **38c**, **38d** is configured to be proximate or adjacent a counter bore **45** of the vehicle part **46** (FIG. 7A) and the side **44b** of each body **38a**, **38b**, **38c**, **38d** is configured to be proximate or adjacent an outer circumferential surface **48** of the vehicle part **46** (FIG. 7B). In the example illustrated, each body **38a**, **38b**, **38c**, **38d** is tapered (FIGS. 2 and 3) from the side **44b** toward the side **44a**. The side **44a** has an arcuate surface **49** and the side **44b** has an arcuate surface **51**. The side **44d** of the body **38a**, **38b**, **38c**, **38d** is opposite the side **44c** and has a flat surface.

Each body **38a**, **38b**, **38c**, **38d** also includes a teeth engagement portion **50** and an attachment portion **52**. When the gear identification tool **34** is on the vehicle part **46**, the teeth engagement portion **50** is located closer toward the outer circumferential surface **48** of the vehicle part **46** than the attachment portion **52**. Each body **38a**, **38b**, **38c**, **38d** includes gear teeth **54** that are configured to engage gear teeth **56** of the vehicle part **46**. The teeth **54** of a corresponding body **38a**, **38b**, **38c**, **38d** are located at the teeth engagement portion **50** at the side **44c** of the corresponding body **38a**, **38b**, **38c**, **38d** (FIG. 4). The teeth **54** of each body **38a**, **38b**, **38c**, **38d** have a predetermined teeth profile that is different from each other and that is configured to mesh with a predetermined teeth profile of teeth of a matching vehicle part. For example, the predetermined teeth profile of the teeth **54** of body **38a** is different from the predetermined teeth profiles of the teeth **54** of bodies **38b**, **38c**, **38d**.

In this way, when the bodies **38a**, **38b**, **38c**, **38d** of the gear identification tool **34** are disposed on the vehicle part **46**, the teeth **54** of the matching teeth profile will mesh and align with the teeth **56** of the vehicle part **46** while the teeth **54** of the other bodies **38** will not mesh or will be misaligned with the teeth **56** of the vehicle part **46**. In the example illustrated, as shown in FIG. 7B, the teeth **54** of body **38d** mesh with the teeth **56** of the vehicle part **46** while the teeth **54** of the other bodies **38a**, **38b**, **38c** are misaligned and/or do not mesh with the teeth **56** of the vehicle part **46** (e.g., see visible gaps between the teeth **54** of the bodies **38a**, **38b**, **38c** and the teeth **56** of the vehicle part **46**).

The teeth profiles of the different bodies **38** may differ in any way or multiple ways, such as pitch, depth, shape,

pressure angle, dimensions, size, thickness, gear diameter, and/or helix angle for example. The teeth profiles may be any suitable type of profile, such as helical, hypoid, spiral, bevel, spur, rack, worm, for example. In the example provided, the teeth profiles are hypoid face gear profiles, though other configurations can be used. In one form, the different bodies **38** may have different types of teeth profiles, such as some bodies having hypoid while others have spiral and others have helical, for example.

When the gear identification tool **34** is on the vehicle part, the arcuate surface **49** of the side **44a** of the matching body **38a**, **38b**, **38c**, **38d** may be flush with a circumferential surface defining a stepped opening of the vehicle part **46**, thereby further assisting the operator in matching the body of the gear identification tool **34** with the vehicle part **46**. The matching body **38a**, **38b**, **38c**, **38d** (i.e., the body **38d** of FIG. 7A) also engages annular land **57b** of the stepped opening **45** whereas a gap may exist between the other bodies **38a**, **38b**, **38c**, **38d** and the annular land **57b** (i.e., a gap exists between the bodies **38a**, **38b**, **38c** of FIG. 7A and the annular land **57b**).

The bodies **38** of the gear identification tool **34** are coupled to each other at the attachment portions **52**. Each body **38a**, **38b**, **38c**, **38d** further includes at least one slot **60a**, at least one aperture **62a** (FIG. 6), and an optional window **63** (FIGS. 2 and 3). The internal slot **60a** formed in each body **38a**, **38d** (i.e., outermost bodies) is located near or at one of the sides **44e**, **44f**. The internal slot **60a** is formed in the bodies **38** at the attachment portion **52**. In the example illustrated, each body **38b**, **38c** (i.e., intermediate bodies) includes at least one additional slot **60b** formed therein. That is, the internal slot **60a** is formed in the body **38b**, **38c** near or at the side **44e** and the internal slot **60b** is formed in the body **38b**, **38c** near or at the side **44f**. Each slot **60a**, **60b** extends in a vertical direction from the side **44c** toward the side **44d**. The aperture **62a** of each body **38a**, **38d** is in communication with the internal slot **60a** and opens through a respective side **44e**, **44f** of the body **38a**, **38d**. In the example illustrated, each body **38b**, **38c** includes an additional aperture **62b** is in communication with the internal slot **60b** and opening through the side **44f** of the body **38b**, **38c**. The side **44f** of one body **38** faces the side **44e** of an adjacent body **38**. For example, the side **44f** of the body **38b** faces the side **44e** of the body **38c**. In the example illustrated, the side **44e** of the bodies **38a**, **38b**, **38c**, **38d** includes a flat surface and the side **44f** of the bodies **38a**, **38b**, **38c**, **38d** includes a flat surface.

In the example illustrated, the optional window **63** is formed in the side **44d** of each body **38** and provides access to the teeth **54** located at the side **44c** of the body **38**. In this way, the operator has visual access of the teeth **54** of the bodies **38a**, **38b**, **38c**, **38d** engaging the teeth of the vehicle part and can visually confirm the body **38a**, **38b**, **38c**, **38d** that mesh with the teeth of the vehicle part. For example, as shown in FIG. 7C, the operator can visually confirm the teeth **54** of the body **38d** meshing with the teeth **54** of the vehicle part **46**. In the example illustrated, the window **63** has a rectangular-shape. In some forms, the window **63** may have a circular shape, square shape, or any other suitable shape that provides visual access of the teeth **54** of the bodies **38a**, **38b**, **38c**, **38d** engaging the teeth of the vehicle part. In some forms, one or more windows (not shown) may be formed in one or more sides **44a**, **44b**, **44e**, **44f** of the body **38**, in addition to, or instead of, the window **63** formed in the side **44d** of the body **38**. In this way, the operator has multiple visual access points of the teeth **54** of the bodies **38**

engaging the teeth of the vehicle part. For example, optional window **65** is formed in the side **44a** of the body **38**.

With reference to FIGS. **5** and **6**, the attachment structures **40** extend between sides **44e**, **44f** of adjacent bodies **38**. For example, the attachment structure **40a** extends between the side **44e** of body **38a** and the side **44f** of the body **38b**. The attachment structures **40** permit the bodies **38** to move in a first direction **X** (FIG. **6**) relative to each other and inhibit the bodies **38** from moving in a second direction **Y** (FIG. **6**) relative to each other that is perpendicular to the first direction **X**. In this way, the bodies **38** of the gear identification tool **34** that do not match with the vehicle part may be misaligned in the first direction **X** relative to the matching body **38**. For example, as shown in FIG. **7A**, the bodies **38a**, **38b**, **38c** are misaligned in the first direction **X** on the vehicle part **46** relative to the matching body **38d**.

With reference to FIGS. **5** and **6**, each attachment structure **40a**, **40b**, **40c** is additively manufactured as a single, monolithic body. In some forms, each attachment structure **40** is manufactured using two or more separate parts that are assembled to each other. Each attachment structure **40a**, **40b**, **40c** includes a first nut **66**, a second nut **68**, and a shaft **70** secured to the first and second nuts **66**, **68**.

The first nut **66** of the attachment structures **40a**, **40b**, **40c** is movably disposed in the internal slot **60a** of a respective body **38a**, **38b**, **38c**. The second nut **68** of the attachment structures **40a**, **40b** is movably disposed in the internal slot **60b** of bodies **38b**, **38c** while the second nut **68** of the attachment structure **40c** is disposed in the internal slot **60a** of the body **38d**. The first nut **66** is disposed in the internal slot **60a** such that the first nut **66** is allowed to move in the first direction **X** in the internal slot **60a** and inhibited from moving in the second direction **Y** in the internal slot **60a**. Similarly, the second nut **68** is disposed in a respective internal slot **60a**, **60b** such that the second nut **68** is allowed to move in the first direction **X** in the respective internal slot **60a**, **60b** and inhibited from moving in the second direction **Y** in the respective internal slot **60a**, **60b**. In this way, the bodies **38** are allowed to move in the first direction **X** relative to each other and inhibited from moving in the second direction **Y** relative to each other.

In the example illustrated, the first and second nuts **66**, **68** have a square shape. In some forms, the first and second nuts **66**, **68** may have a circular shape, an octagonal shape, or any other suitable shape that permits movement in the first direction **X** within the slot while inhibiting movement in the second direction **Y** within the slot. The shaft **70** securing the first and second nuts **66**, **68** to each other inhibits the bodies **38** from moving in the second direction **Y** relative to each other. The shaft **70** also extends through the apertures **62a**, **62b** of adjacent bodies **38a**, **38b**, **38c**, **38d**. It should be understood that although the present disclosure provides the gear identification tool **34** for identifying vehicle parts, the gear identification tool **34** may also be used for identifying other type of parts such as parts of machinery, for example.

With reference to FIG. **8**, another gear identification tool **134** is provided. The structure and function of the gear identification tool **134** may be similar or identical to that of gear identification tool **34** described above, apart from any exception noted below. In the example illustrated, the gear identification tool **134** is a segmented assembly having a circular profile and including a plurality of bodies **138a**, **138b**, **138c**, **138d** (together referred to as the bodies **138**) and attachment structures **140a**, **140b**, **140c**, **140d** (together referred to as the attachment structures **140**).

The attachment structures **140** permit the bodies **138** to move in a radial direction relative to each other and inhibit

the bodies **138** from moving in a axial direction relative to each other. In this way, the bodies **138** of the gear identification tool **134** that do not match with a vehicle part **150** (e.g., a splined shaft) are misalign on the vehicle part **150**.

Unless otherwise expressly indicated herein, all numerical values indicating mechanical/thermal properties, compositional percentages, dimensions and/or tolerances, or other characteristics are to be understood as modified by the word “about” or “approximately” in describing the scope of the present disclosure. This modification is desired for various reasons including industrial practice, material, manufacturing, and assembly tolerances, and testing capability.

As used herein, the phrase at least one of A, B, and C should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean “at least one of A, at least one of B, and at least one of C.”

The description of the disclosure is merely exemplary in nature and, thus, variations that do not depart from the substance of the disclosure are intended to be within the scope of the disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the disclosure.

What is claimed is:

1. A gear identification tool comprising:

a segmented assembly including a first body and a second body movable relative to each other, the first and second bodies coupled to each other by a first attachment structure, the first body including a plurality of first gear teeth configured to mesh with first predetermined gear teeth of a first gear, wherein the first gear teeth have a first teeth profile, the second body including a plurality of second gear teeth configured to mesh with second predetermined gear teeth of a second gear, wherein the second gear teeth have a second teeth profile that is different than the first teeth profile.

2. The gear identification tool of claim 1, wherein:

the first body defines a first internal slot and a first aperture in communication with the first internal slot, the first aperture opens through a first side of the first body; and

the second body defines a second internal slot and a second aperture in communication with the second internal slot, the second aperture opens through a second side of the second body.

3. The gear identification tool of claim 2, wherein the first attachment structure extends between the first side of the first body and the second side of the second body.

4. The gear identification tool of claim 1, wherein the first attachment structure permits the first and second bodies to move in a first direction relative to each other and inhibits the first and second bodies from moving in a second direction relative to each other, the second direction being perpendicular to the first direction.

5. The gear identification tool of claim 1, wherein the first attachment structure permits the first and second bodies to move within a predetermined distance relative to each other.

6. The gear identification tool of claim 1, wherein the first attachment structure includes:

a first nut located in a first internal slot of the first body; a second nut located in a second internal slot of the second body; and

a shaft secured to the first and second nuts.

7. The gear identification tool of claim 6, wherein the first attachment structure is a unitized structure.

8. The gear identification tool of claim 1, wherein the segmented assembly includes a third body coupled to the

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second body by a second attachment structure, and wherein the third body is permitted to move relative to the first and second bodies.

9. The gear identification tool of claim 8, wherein the third body includes a plurality of third gear teeth configured to mesh with third predetermined gear teeth of a third gear, the third gear teeth having a third teeth profile that is different than the first and second teeth profiles.

10. The gear identification tool of claim 1, wherein the first body has a first side including the first gear teeth and a second side opposite the first side, the second side comprising a window that provides access to the first teeth.

11. The gear identification tool of claim 1, wherein the segmented assembly has an arcuate profile disposed about a central axis, the first body extending a first angle relative to the central axis and the second body extending a second angle relative to the central axis.

12. The gear identification tool of claim 1, wherein the first and second gear teeth are face gear teeth.

13. A gear identification tool comprising:

a segmented assembly including a plurality of bodies coupled to each other by attachment structures and movable relative to each other, each body including a plurality of gear teeth configured to mesh with predetermined gear teeth of a respective gear;

wherein each body includes an internal slot and an aperture in communication with the internal slot, the aperture opens through a first side of the body, and

wherein one attachment structure of the attachment structures includes a first nut located in the internal slot of one body of the plurality of bodies, a second nut located in the internal slot of another body of the plurality of bodies, and a shaft secured to the first and second nuts.

14. The gear identification tool of claim 13, wherein the gear teeth of the one body of the plurality of bodies have a first teeth profile and the gear teeth of the another body of the plurality of bodies have a second teeth profile, and wherein the first gear teeth profile is different than the second teeth profile.

15. The gear identification tool of claim 13, wherein the one attachment structure permits the one body and the another body to move within a predetermined distance relative to each other.

16. The gear identification tool of claim 13, wherein the one attachment structure permits the one body and the

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another body to move in a first direction relative to each other and inhibits the one body and the another body from moving in a second direction relative to each other, the second direction being perpendicular to the first direction.

17. The gear identification tool of claim 13, wherein the one body has a second side including the gear teeth and a third side opposite the second side, and wherein the third side comprises a window that provides access to the gear teeth.

18. The gear identification tool of claim 13, wherein the segmented assembly has an arcuate profile disposed about a central axis, the one body extending a first angle relative to the central axis and the another body extending a second angle relative to the central axis.

19. The gear identification tool of claim 13, wherein the segmented assembly includes at least four bodies.

20. A gear identification tool comprising:

a segmented assembly including:

a plurality of bodies movable relative to each other, each body including a plurality of gear teeth having a gear teeth profile that is different from the gear teeth profile of the other bodies of the plurality of bodies, each body also including an internal slot and an aperture in communication with the internal slot, the aperture opens through a first side of the body; and

attachment structures coupling the plurality of bodies to each other, one attachment structure of the attachment structures includes a first nut located in the internal slot of one body of the plurality of bodies, a second nut located in the internal slot of another body of the plurality of bodies, and a shaft secured to the first and second nuts,

wherein the one attachment structure permits the one body and the another body to move in an axial direction relative to each other and inhibits the one body and the another body from moving in a radial direction relative to each other, and

wherein each of the plurality of bodies has a second side including the gear teeth and a third side opposite the second side, and wherein the third side comprises a window that provides access to the gear teeth.

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